A Novel Arctic Cloud Phase Classifier Based on Ship-Based Remote Sensing Measurements During the MOSAiC Expedition

Jin Ye, Lei Liu, Hailing Xie, and Meihua Wang

***Abstract*— Accurately measuring the phase of Arctic clouds is of great significance for understanding the process of Arctic climate change. A novel algorithm for classifying Arctic cloud phase is constructed based on ship-based remote sensing measurements during the Multidisciplinary drifting Observatory for the Study of Arctic Climate (MOSAiC) expedition. Based on complementary information by Ka-band ARM zenith radar (KAZR), high spectral resolution lidar (HSRL), microwave radiometer (MWR), and temperature soundings, cloud pixels in the vertical direction can be classified as clear sky, aerosol, ice, mixed phase, liquid, drizzle, rain, and snow. The dataset collected from September 2019 to September 2020 is analyzed. Results indicate that the proposed method can effectively distinguish Arctic mixed-phase clouds (MPCs) with low depolarization ratio from liquid clouds. Even if the MPCs contain the lower concentration of ice crystals, the proposed algorithm can accurately identify them. Simultaneously, the cloud phase classification is less affected by environmental factors. It provides a reference for subsequent cloud microphysical retrievals, reducing significant deviations in droplet microphysical retrievals caused by misidentifying MPC as liquid cloud. The algorithm proposed in this article makes appropriate modifications to cloud classification boundaries and can also be applied to other sites.**

***Index Terms*— Arctic cloud phase classification, lidar, microphysical property, Multidisciplinary drifting Observatory for the Study of Arctic Climate (MOSAiC), radar, vertical structure.**

I. Introduction

In RECENT decades, the Arctic has undergone unprecedented changes, and the causes of these changes and their feedback to the global climate system are currently under intensive research [1]. The rate of warming in the Arctic is faster than any other region on Earth [2], a phenomenon known as Arctic amplification [3]. Many studies have elaborated on the significant impact of Arctic clouds, especially low-level mixed-phase clouds (MPCs), on the surface radiation budget [2], [4], [5], [6], which is closely related to the growth and melting processes of sea ice [7], [8]. They regulate the radiation budget of the Earth's surface by reflecting or absorbing solar radiation, and participate in the evaporation, transport, and precipitation processes of water, thereby controlling the stability and variability of climate on a global scale [9], [10], [11].

Previous studies have shown that MPCs are ubiquitous, being one of the most common, longest lasting, and most radiatively influential clouds in the Arctic [12], [13], [14]. Observations indicate that Arctic MPCs can persist for tens of hours or even days under various meteorological conditions [15], [16], [17], often occurring alongside the emergence of supercooled clouds, typically beginning shortly after the formation of supercooled clouds [18]. Due to the lack of adequate descriptions of numerous microphysical and dynamical processes within MPCs, global climate models (GCMs) significantly overestimate the contribution of Wegener–Bergeron–Findeisen (WBF) processes to the consumption of supercooled water [11]. It leads to a significant underestimation of the frequency of polar supercooled and MPCs in GCMs [19], [20] and reanalysis data [21].

The cloud thermodynamic phase is an important parameter due to the crucial role in the clouds' radiative effect [5]. The distribution between the content of liquid droplets and ice crystals determines the optical depth and lifecycle of the clouds [22]. Moreover, accurately obtaining the phase of Arctic clouds is a prerequisite for retrieving cloud microphysical parameters from remote sensing measurements [23]. However, accurately retrieving the phase distribution of Arctic MPCs and linking it to environmental conditions (altitude, temperature, surface conditions, etc.) for parameterization and modeling remains a significant challenge [22]. Incorrect identification of cloud phase may lead to substantial errors in the retrieval of cloud microphysical parameters.

Satellites can provide significant spatial and temporal coverage for observing polar clouds [1], [24], [25]. But due to the high reflectivity of the Arctic surface, it is difficult to distinguish between the underlying surface and polar low-level clouds based on satellite passive remote sensing [26]. For satellite active remote sensing, however, due to its low vertical resolution and susceptibility to ground clutter (such as challenges in distinguishing fog and low clouds), the retrieved Arctic cloud phase may have significant uncertainty [27], [28]. To accurately obtain the phase of Arctic clouds, some joint

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ground-based multisensor approaches have been developed for cloud phase identification. This is an important supplement to satellite remote sensing. For example, Shupe proposed an Arctic cloud classifier based on cloud radar, lidar, microwave radiometer (MWR), and temperature soundings, which distinguishes between liquid cloud and MPC based on radar reflectivity, Doppler velocity, and spectral width (SW) [23]. The Cloudnet cloud phase classification algorithm similarly uses the vertical velocities of targets observed by radar to identify falling ice particles in temperatures below 0 °C, thereby distinguishing between liquid cloud and MPC [29], [30], [31]. Silber et al. [32] used a single-frequency zenith-pointing radar to classify the Arctic cloud phase and found that the SW is the radar moment with the highest discrimination between liquid and ice. However, since these algorithms involve SW and Doppler velocity, the retrieval results may be affected by atmospheric vertical motion states, turbulence, and wind shear [33]. Kalogeras et al. [33] examined the climatology of the North Slope of Alaska (NSA) site over a period of 5.5 years and found that SW clustering around 0.1 m/s in supercooled clouds. The turbulence inside the supercooled clouds was relatively small, and the corresponding SW rarely exceeded 0.25 m/s. However, due to the enhanced turbulence at the top of supercooled clouds, the SW of the condensation zone at the top of supercooled clouds can rise to over 0.25 m/s, and even exceed 0.3 m/s [33], [34].

The Multidisciplinary drifting Observatory for the Study of Arctic Climate (MOSAiC) expedition was designed to collect a wealth of data in the Arctic Ocean from September 2019 to September 2020 [35], [36]. MOSAiC expedition deployed with Ka-band ARM zenith radar (KAZR) [37], high spectral resolution lidar (HSRL) [38], MWR [39], etc. It can provide fine-grained remote sensing of clouds over the Arctic Ocean. To effectively reduce the impact of environmental factors on the classification of Arctic cloud phase, based on the algorithm developed by Shupe [23], we propose a novel Arctic cloud phase classifier based on HSRL, KAZR, MWR, and temperature soundings [40] in this article. To better distinguish the phase for areas with less turbulence inside the cloud, we construct the relationship between the reflectivity of KAZR and the extinction coefficient of HSRL. The structure of this article is as follows. Section II introduces the data involved in this work; Section III summarizes the specific steps of cloud phase classification algorithm; Section IV further verifies the accuracy of the proposed algorithm; Finally, the conclusion are presented in Section V.

II. Data and Instruments

A. Instruments on Polarstern

The MOSAiC expedition is a year-long expedition aimed at studying Arctic climate change and its connection to the global system [35], [41]. It began in September 2019 in the central Arctic (north of the Laptev Sea), where the German research icebreaker *Polarstern* [42] was frozen into the sea ice. The *Polarstern* carried about various instruments for continuous observation of Arctic clouds and drifted with sea ice for most of the next 13 months across the transpolar drift [43], [44], [45]. The entire expedition was divided into five legs by different participants, namely, Leg 1—Installation, Leg 2—Winter, Leg 3—Rising Sun, Leg 4—Summer Melt, and Leg 5—Freeze Up [36]. In this article, the thermodynamic cloud phase value-added product (THERMOCLDPHASE) is used [46], [47], which combines KAZR, HSRL, and MWR. Meanwhile, Interpolated Sounding and Gridded Sounding Value-Added Products (INTERPOLATEDSONDE) is also included, which can provide temperature, humidity, and wind profiles [40]. KAZR operates continuously in an alternating fashion between two distinct modes: the general (GE) mode and the medium sensitivity (MD) mode [48]. Using the radar mode flag, THERMULDPHASE is capable of determining the optimal radar operating mode. Therefore, THERMOCLDPHASE can provide best estimate radar reflectivity, mean Doppler velocities, and SW of KAZR. Besides, it can also provide backscattering coefficient, depolarization ratio, and extinction coefficient profile of HSRL with a vertical resolution of 30 m from 0.16 to 18 km every minute. The uncertainty of the liquid water path (LWP) of MWR products is 25 $g \cdot m^{-2}$, which is interpolated to a 1-min resolution to match the radar time resolution [39], [49], [50]. However, when the rocking angle (includes pitch and roll angles) of the *Polarstern* is not equal to zero, it causes the nonvertical radar beam pointing. Therefore, we further match the *Polarstern*'s navigation information [51].

B. Instruments on Polar 5

The airborne campaigns atmospheric airborne observations in the central Arctic (ACA) conducted by the research aircraft *Polar 5* complemented the Arctic cloud remote sensing results obtained by the German research vessel *Polarstern* during the MOSAiC drift across the Arctic Ocean [35], [52]. Due to the impact of the COVID-19 pandemic, MOSAiC-ACA has canceled its spring sports. The summer campaigns were held from August 30 to September 17, 2020 in Longyearbyen, Svalbard (78°13′N, 15°38′E), covering a wide geographical area from open ocean to sea ice covered regions (ten measurement flights) [53]. *Polar* 5 is equipped with in situ, remote sensing, and basic instruments such as the 2-D stereo cloud probe, the cloud combination probe, lidar, Sun photometer, hyperspectral camera, spectral solar radiometer, cloud radar systems, and standard meteorological sensors [35], [53].

III. Algorithm Construction

In this section, we analyze the distribution of microphysical parameters of liquid clouds observed during MOSAiC-ACA and calculate the geometric standard deviation of droplet size. Ulteriorly, we construct the Arctic cloud phase classification algorithm based on ship-based remote sensing measurements.

A. Geometric Standard Deviation of Droplet Size

Due to the fact that the cloud phase classification algorithm in Section III-B applies a constant geometric standard deviation of droplet size (σ_g), we analyze that characteristic based on *Polar* 5 field measurement. Similar to the airborne field experiments in the Southeast Pacific by Snider et al. [54],

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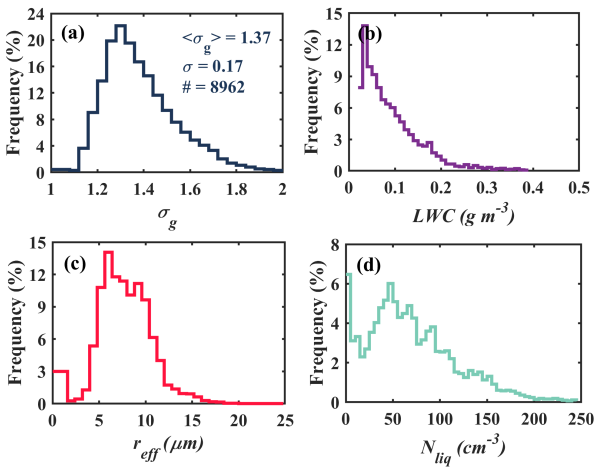


Fig. 1. Microphysical parameters from the in situ cloud measurements carried by *Polar 5*. (a) Frequency of σ_g based on 1-s measurements. Properties of the dataset (mean, standard deviation, and number of values) are display in the legend. Frequency of (b) LWC, (c) droplet effective radius, and (d) droplet concentration corresponding to liquid cloud.

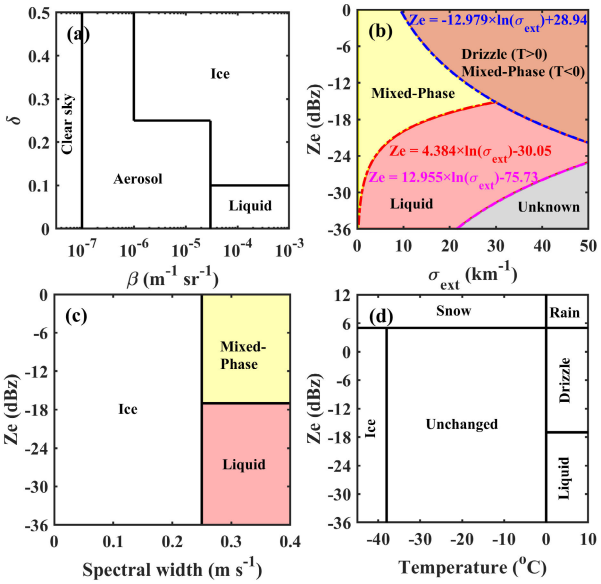


Fig. 2. Cloud phase classification diagram. (a) Preliminary classification of cloud phase based on the backscattering coefficient and depolarization ratio from HSRL. (b) Correction results of liquid cloud using the reflectivity of KAZR and the extinction coefficient of HSRL. (c) Classification of cloud pixels undetected by HSRL using KAZR reflectivity and SW. (d) Using temperature profiles and KAZR reflectivity to further correct cloud phase.

we select the clouds with liquid water content (LWC) greater than 0.02 g · m^{-3} measured by the Nevzorov bulk probe during MOSAiC-ACA. To avoid interference from ice crystals on the calculation of geometric standard deviation of droplet size, we further add the condition that the liquid fraction exceeds 90% to determine the liquid cloud. The liquid fraction is defined as the ratio of LWC to total water content (TWC) [13]. σ_g can be calculated as follows [55]:

$$\ln \sigma_g = \left[\frac{\sum n_i \left(\ln r_i - \ln r_g \right)^2}{N - 1} \right]^{1/2} \tag{1}$$

where r_i represents the midpoint particle radius of the ith bin; n_i represents the number of particles in group i with a midpoint size r_i, which can be measured by cloud droplet probe (CDP); and N represents the total number of particles. r_g represents the geometric mean radius, which can be calculated as follows:

$$r_g = \left(r_1^{n_1} r_2^{n_2} r_3^{n_3} \cdots r_N^{n_N} \right)^{1/N}. \tag{2}$$

The distribution range of σ_g is shown in Fig. 1(a). The mean of σ_g is 1.37, and the majority of σ_g falls within the range of 1.1–1.8. We find that σ_g during MOSAiC-ACA is similar to the results calculated by Snider et al. [54] in the airborne field experiments over Southeast Pacific. In addition, Fig. 1 shows the distribution of microphysical parameters of liquid clouds during MOSAiC-ACA. The LWC is generally less than 0.3 g · m^{-3}, the droplet effective radius (r_{eff}) does not exceed 15 μm, and the droplet concentration (N_{liq}) is within 200 cm^{-3}, which is consistent with existing research on Arctic supercooled clouds [1], [22], [27].

B. Cloud Phase Classification Algorithm

According to the cloud phase classification algorithm developed Shupe [23], we propose a novel Arctic cloud phase classification algorithm base on KAZR (reflectivity, SW), HSRL (depolarization ratio, extinction coefficient, backscattering coefficient), MWR (LWP), and temperature profiles. The specific steps of the proposed algorithm are as follows.

Step 1: A preliminary classification of cloud phase is conducted based on the backscattering coefficient and depolarization ratio from HSRL [see Fig. 2(a)]. Similar to Shupe's method, here cloud pixels observed by HSRL are categorized into clear sky, ice, aerosol, and liquid. Among them, liquid cloud is characterized by high backscattering and low depolarization ratio, which is composed by high concentrations of small droplets. In contrast, the common low concentration of irregular ice particles in the Arctic exhibits low backscattering and high depolarization ratio [56].

Step 2: When no liquid cloud is detected in Step 1, and if the LWP exceeds the uncertainty (25 g · m^{-2}), classify the ice cloud with a high backscattering coefficient (greater than 3 × 10^{-5} m^{-1} · sr^{-1}) as a liquid cloud.

Step 3: The classification results of liquid cloud in Steps 1 and 2 are corrected using the reflectivity (Ze) of KAZR and the extinction coefficient (σ_{ext}) of HSRL [see Fig. 2(b)]. In this article, we assume that the droplet size during MOSAiC follows a lognormal distribution with a constant geometric standard deviation [27]. We have analyzed the distribution of σ_g in Section III-A, and here we select the constant σ_g as 1.37. The σ_{ext} and LWC can be expressed as

$$\sigma_{\text{ext}} = 2\pi N_{\text{liq}} \langle r^2 \rangle = 2\pi N_{\text{liq}} r_0^2 e^{2 \ln^2(\sigma_g)} \tag{3}$$

$$\text{LWC} = \frac{4}{3} \pi \rho_w N_{\text{liq}} \langle r^3 \rangle = \frac{4}{3} \pi \rho_w N_{\text{liq}} r_0^3 e^{9 \ln^2(\sigma_g)/2} \tag{4}$$

where ρ_w represents the density of water; r_0 represents the modal radius; and $< r^k >$ represents the kth moment of the droplet size distribution. For a lognormal distribution, $< r^k >$

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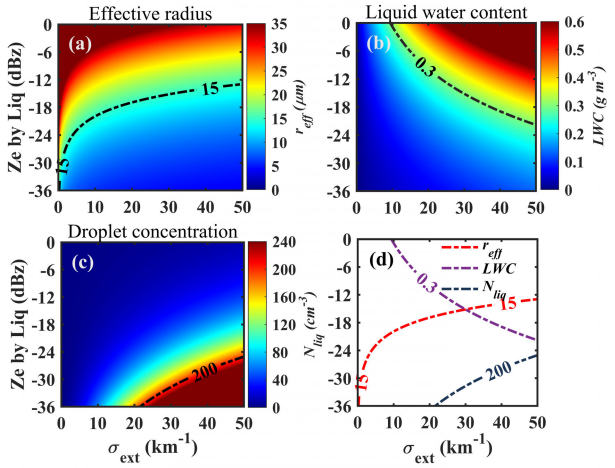


Fig. 3. Relationships between the reflectivity and the extinction coefficient with (a) droplet effective radius, (b) LWC, and (c) droplet concentration, as calculated based on theoretical formulas. The black line in the figure represents the extreme values of droplet microphysical parameter distribution of Arctic liquid cloud. (d) Boundaries of the three extrema corresponding to droplet microphysical parameters.

can be expressed as

$$\langle r^k \rangle = r_0^k e^{k^2 \ln^2(\sigma_g)/2} \tag{5}$$

where r_{eff} is the ratio of the 3rd moment to the 2nd moment of the droplet size distribution as follows:

$$r_{\text{eff}} = \langle r^3 \rangle / \langle r^2 \rangle = r_0 e^{5 \ln^2(\sigma_g)/2}. \tag{6}$$

The reflectivity Ze received by the radar is a function of the 6th moment of the droplet size distribution as follows:

$$Ze = 2^6 N_{\text{liq}} \langle r^6 \rangle = 2^6 N_{\text{liq}} r_0^6 e^{18 \ln^2(\sigma_g)}. \tag{7}$$

By connecting (3)–(7), the relationships between the reflectivity and the extinction coefficient with r_{eff}, LWC, and N_{liq} can be obtained as shown in Fig. 3. Section III-A shows that r_{eff} in supercooled clouds is generally less than 15 μm, the LWC does not exceed 0.3 g/m^3, and N_{liq} is no more than 200 cm^{-3} (as shown by the black line in Fig. 3). We further parameterize the boundaries of the three extrema in Fig. 2(b)

$$Ze = 4.384 \times \ln(\sigma_{\text{ext}}) - 30.05 \tag{8}$$

$$Ze = -12.979 \times \ln(\sigma_{\text{ext}}) + 28.94 \tag{9}$$

$$Ze = 12.955 \times \ln(\sigma_{\text{ext}}) - 75.73. \tag{10}$$

Equations (8)–(10), respectively, represent the relationship between reflectivity (unit: dBZ) and extinction coefficient (unit: km^{-1}) corresponding to the extreme values of r_{eff}, LWC, and N_{liq}. Based on the distribution of extreme values of droplet microphysical parameters, Fig. 2(b) further corrects the phase of liquid cloud identified in Steps 1 and 2. When the extinction coefficient of HSRL and reflectivity of KAZR are located in the yellow area of Fig. 2(b), liquid cloud is reclassified as MPC; when the extinction coefficient and reflectivity are located in the orange area and the temperature is less than (more than) 0 °C, it is reclassified as MPC (drizzle); when the extinction coefficient and reflectivity are located in the gray area, it is corrected to unknown. In addition, if the extinction coefficient is greater than 50 km^{-1}, it is also defined as unknown, possibly due to noise or other factors causing anomalies in the extinction coefficient.

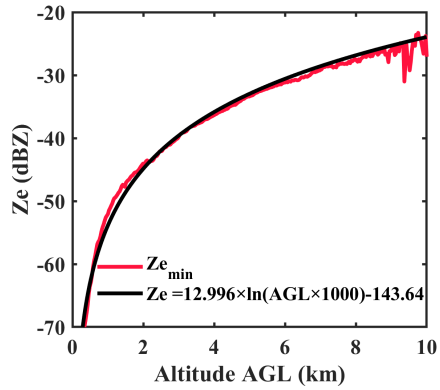


Fig. 4. Minimum threshold for detecting cloud pixels at each altitude using KAZR reflectivity (red line). The black line represents the parameterization result for the minimum threshold.

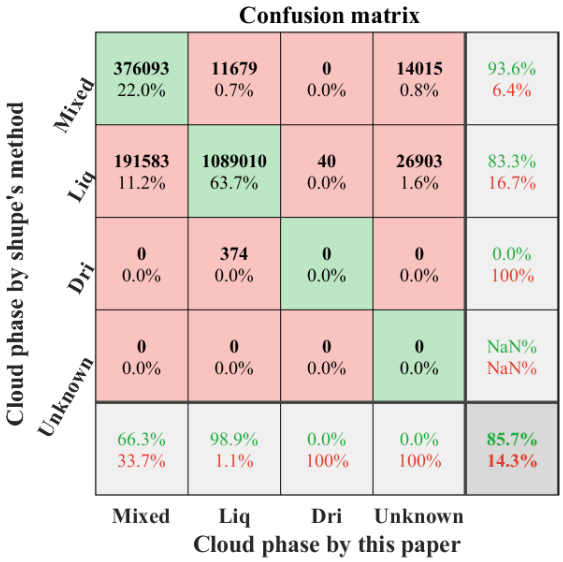


Fig. 5. Confusion matrix between the proposed algorithm and Shupe's method for identifying liquid and MPCs.

Step 4: Use KAZR reflectivity and SW to classify the cloud pixels undetected by HSRL [see Fig. 2(c)]. When the Arctic cloud layer is thick, the corresponding beam of HSRL cannot penetrate through. Moreover, HSRL backscattering is not sensitive to low-concentration ice clouds, and such ice clouds cannot be identified from the HSRL backscattering signal. Therefore, it is necessary to use KAZR to supplement the HSRL cloud detection results in Step 1. We use a ceilometer to find the clear-sky moments during MOSAiC expedition and calculate the mean (μ_{Ze}) and standard deviation (σ_{Ze}) of KAZR reflectivity at each altitude (above ground level, AGL) for the corresponding moments. Here, the minimum thresholds ($Ze_{\min}$) for detecting cloud pixels at each altitude based on reflectivity are defined by $\mu_{Ze}^{-3} \times \sigma_{Ze}$ [57], [58], which is shown by the red line in Fig. 4. Meanwhile, we also parameterize $Ze_{\min}$ shown by the black line.

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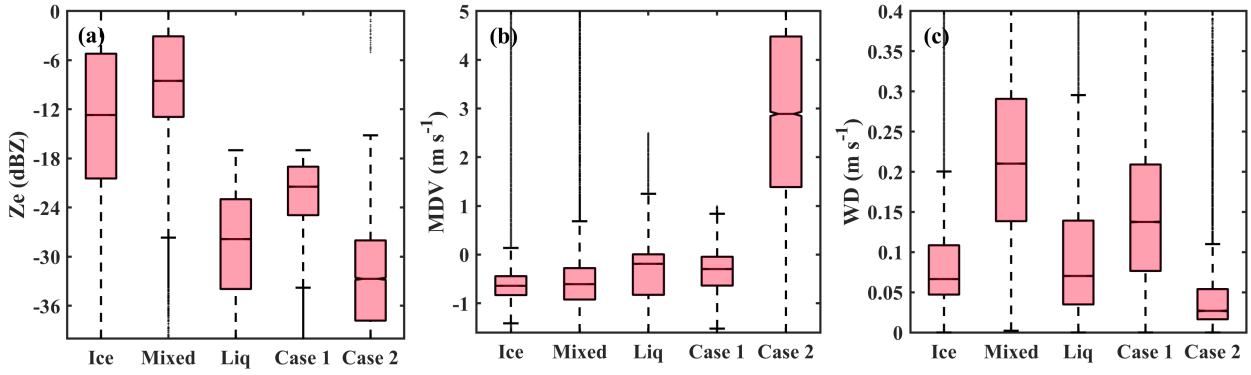


Fig. 6. Distribution of (a) reflectivity, (b) MDV, and (c) SW for two algorithms simultaneously identified as ice cloud, MPC, liquid cloud, Case 1, and Case 2.

When cloud pixels are not detected by HSRL, but the corresponding KAZR reflectivity is greater than $Ze_{\min}$, we mark these pixels as ice cloud first. If the signal emitted by HSRL is completely blocked in the cloud, it will cause the liquid cloud top to be underestimated. Due to the significant enhanced turbulence in the condensation zone of Arctic supercooled cloud top, we further use radar reflectivity and SW to identify supercooled cloud top. Based on the 5.5-year climatological statistics of NSA site by Kalogeras et al. [33], we select an SW threshold of 0.25 m $\cdot$ s^{-1} for supercooled cloud detection. Meanwhile, as shown in Fig. 2(b), the reflectivity corresponding to liquid clouds is all less than -17 dBZ. Therefore, when the HSRL's signal is completely blocked upon reaching the top of the liquid layer, if the KAZR reflectivity is between $Ze_{\min}$ and -17 dBZ (greater than -17 dBZ), and the SW is more than 0.25 m $\cdot$ s^{-1}, these parts of the ice pixels are reclassified as liquid (mixed phase) [59].

Step 5: Using temperature profiles and KAZR reflectivity to correct cloud phase [see Fig. 2(d)]. -38 °C is the limit temperature for the existence of liquid water [45]. So, for MPCs and supercooled clouds, if the corresponding temperature is lower than -38 °C, it should be reclassified as ice cloud. While the temperature is above 0 °C, only liquid hydrometeors are allowed to exist. In this case, if the KAZR reflectivity is over (below) -17 dBZ, it is reclassified as drizzle (liquid cloud).

Step 6: Using LWP of MWR to constrain cloud phase. If the LWP ≤ 0 g $\cdot$ m^{-2}, it indicates that there are no droplets present in the cloud. Furthermore, if T $<$ 0 °C, all liquid-containing classifications in the vertical column are converted into ice.

Step 7: Using temperature profiles and KAZR reflectivity to identify rain and snow [see Fig. 2(d)]. While the KAZR reflectivity is greater than 5 dBZ, it is reclassified as rain (T $>$ 0 °C) or snow (T $<$ 0 °C). These criteria apply to all cloud pixels.

IV. Algorithm Evaluation

The algorithm proposed in this article and Shupe's method largely share the same basis for judging ice cloud, snow, and rain. Therefore, this section primarily compares the classification results of liquid and MPCs between the two algorithms. Fig. 5 presents a comparison of the results identified as liquid or MPCs by either the proposed method or Shupe's method. From the confusion matrix, it can be seen that the two methods have good consistency in identifying liquid and MPCs, approximately 85.7%. However, the proposed method tends to classify cloud pixels as mixed phase, while Shupe's method is more inclined to identify them as liquid cloud. It can be observed that there are quite a few cloud pixels (about 11.2%) that the proposed method judges as mixed phase, whereas Shupe's method judges them as liquid. Similarly, it is found that about 0.7% of cloud pixels are judged by Shupe's method as mixed phase, while the proposed method judges them as liquid. Moreover, there are about 2.4% of cloud pixels that the proposed method judges as undefined, corresponding to an abnormally large extinction coefficient of HSRL, often occurring near positions where HSRL's signal is completely obscured by clouds.

To further analyze the classification accuracy of the two methods, we classify cloud pixels with inconsistent classifications into two cases.

Case 1: The proposed method classifies as mixed-phase cloud, while Shupe's method classifies as liquid cloud.

Case 2: The proposed method classifies as liquid cloud, while Shupe's method classifies as MPCs.

Fig. 6 presents the distribution of reflectivity, mean Doppler velocity (MDV), and SW for two algorithms simultaneously identified as ice cloud, MPC, liquid cloud, Case 1, and Case 2. Consistent with the observation results of Kalogeras et al. [33], liquid clouds exhibit an SW clustering around 0.1 m s^{-1}. During the MOSAiC, the MDVs are mostly negative, indicating that ascending airflow was dominant. Only Case 2 exhibits larger positive MDV with smaller reflectivity, which led Shupe's method to classify it as an MPC. It is important to note that when the roll or pitch angle is large and there are strong horizontal winds, this can result in a substantial Doppler velocity, which will affect the accuracy of Shupe's method. Notably, Case 1, which accounts for a larger proportion, exhibits distribution characteristics similar to those of liquid clouds in terms of reflectivity, MDV, and SW. We further compare the accuracy of the two algorithms in detail under Cases 1 and 2 in Sections IV-A and IV-B. At the same time, we have also analyzed the impact of ship shaking on the cloud

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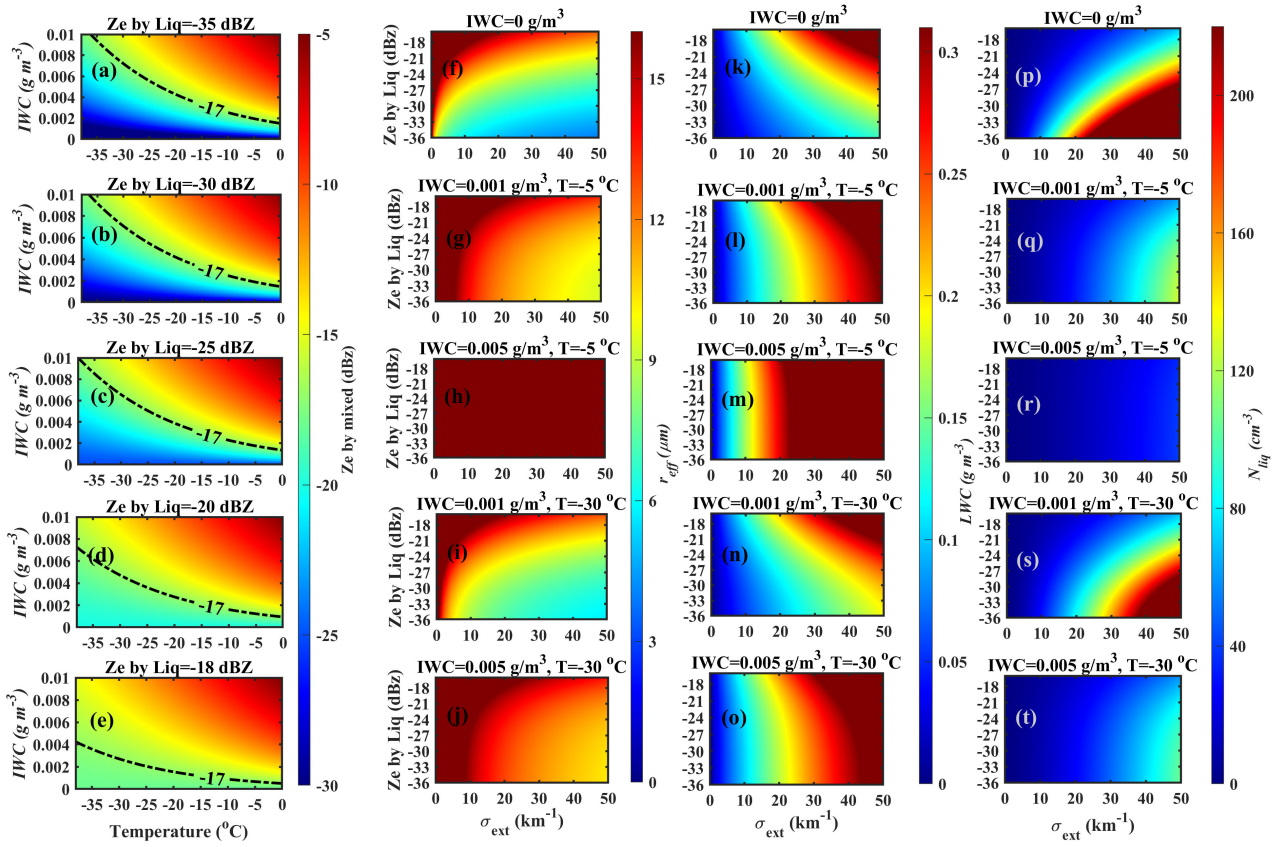


Fig. 7. Simulation results of the relationship between different microphysical parameters in MPC and reflectivity and extinction coefficient. (a)–(e) Contribution of different liquid clouds to radar reflectivity in MPCs and further superimpose different IWCs at different temperatures. Color bar represents the total reflectivity corresponding to the MPC. The impact of different IWCs at various temperatures on the inversion of (f)–(j) droplet effective radius, (k)–(o) LWC, and (p)–(t) droplet concentration, which are calculated based on the reflectivity corresponding to MPCs.

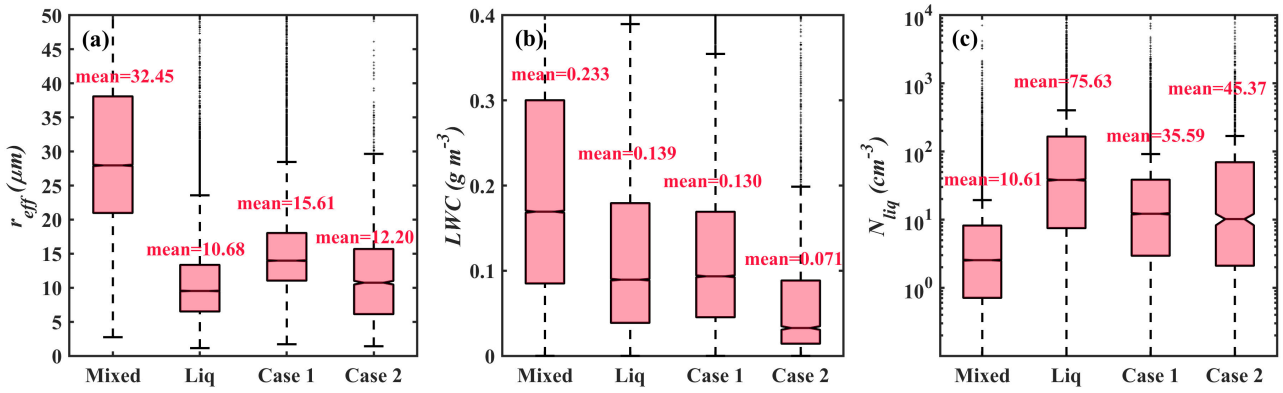


Fig. 8. Distribution of (a) droplet effective radius, (b) LWC, and (c) droplet concentration corresponding to MPC, liquid cloud, Case 1, and Case 2 using KAZR and MWR measurement data.

classification outcomes produced by the proposed algorithm, as detailed in the Appendix.

A. Impact of IWC in MPCs on Cloud Phase Classification

From Fig. 6, it can be seen that the reflectivity of −17 dBZ plays a crucial role in distinguishing liquid and MPC. To understand the contribution of ice crystals with different ice water contents (*IWCs*) to the reflectivity of MPCs, further analysis of the impact of IWC in MPC on the result of cloud phase classification is conducted. In Fig. 7(a)–(e), we simulate the contribution of different liquid clouds to radar reflectivity (Ze_{liq}) in MPCs and further superimpose different *IWCs* at different temperatures. The contribution of ice crystals to radar reflectivity (Ze_{ice}) can be calculated using Ze_{ice}–IWC–T relationship proposed by Hogan et al [60]. The color bar in the figure represents the reflectivity (Ze_{mixed}) of MPC, which can be calculated using

$$Ze_{\mathrm{mixed}} = 10 \times \log_{10}\left(10^{0.1\times Ze_{\mathrm{liq}}} + 10^{0.1\times Ze_{\mathrm{ice}}}\right). \tag{11}$$

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Ze_{liq}, Ze_{ice}, and Ze_{mixed} are measured in dBZ, IWC is measured in g · m^{-3}, and temperature T is measured in °C. Fig. 7(a)–(e) shows that when the IWC in MPCs is small, it does not lead to significant changes in radar reflectivity. Furthermore, the extent of this change decreases with the lowering of temperature, but Shupe's method does not identify it as an MPC. Only when the IWC and temperature are high enough to make the radar reflectivity of the MPC greater than −17 dBZ will the Shupe's method identify it as an MPC.

However, even ice crystals with a small IWC can significantly interfere with the inversion results of MPC microphysical parameters. Fig. 7(f)–(t) shows the impact of different *IWCs* at various temperatures on the inversion of droplet microphysical parameters, which are calculated based on the reflectivity corresponding to MPCs. It is evident that when MPCs are mistakenly identified as liquid clouds, the obtained droplet microphysical parameters will deviate significantly with the increase in IWC and temperature. Overall, as the IWC increases, both r_{eff} and the LWC obtained will obviously overestimate, while the corresponding N_{liq} will substantially underestimate. For example, in Fig. 7(g), at a temperature of −5 °C, even IWC only with 1 mg · m^{-3} can lead to a significant overestimation of the effective radius of droplets (r_{eff} over 15 μm). Based on the algorithm in this article, it would be identified it as MPC. But combined with Fig. 7(a)–(e), we find that such ice crystals cause the reflectivity of MPC to be less than −17 dBZ, and Shupe's method might identify it as liquid cloud.

B. Algorithm Verification Based on KAZR and MWR Measurement Data

Based on Section IV-A, which discusses the impact of ice and LWC in MPCs on cloud phase classification, we further use KAZR and MWR measurement data to retrieve the microphysical parameters of droplets corresponding to different cases (Refer to [59]). Consistent with Fig. 7(f)–(t), this section does not separate the contribution of ice crystals to radar reflectivity. We directly use the reflectivity of KAZR and the LWP of MWR corresponding to MPCs for the retrieval of droplet microphysical parameters. As observed in Section IV-A, when the obtained r_{eff} and LWC are larger or N_{liq} is smaller, it indicates that the corresponding cloud pixels contain ice crystals.

Fig. 8 shows the distribution of droplet microphysical parameters corresponding to MPC, liquid cloud, Case 1, and Case 2. Consistent with the conclusions from the simulation results in Section IV-A, misidentifying MPC as liquid cloud will lead to significant deviations in the droplet microphysical retrievals, with r_{eff} almost always between 20 and 40 μm. In contrast, for liquid cloud, r_{eff} is almost entirely less than 15 μm, with an average r_{eff} of 10.68 μm. In addition, the LWC mostly ranges between 0.05 and 0.2 g m^{-3}, and N_{liq} is between 10 and 150 cm^{-3}. This aligns with the classification criteria shown in Fig. 2(b).

In addition, we can find that r_{eff} inversion results of Case 1 slightly larger than those of liquid clouds. The r_{eff} in Case 1 ranges from 12 to 18 μm, with an average r_{eff} of 15.61 μm. Moreover, N_{liq} is slightly less than that of liquid

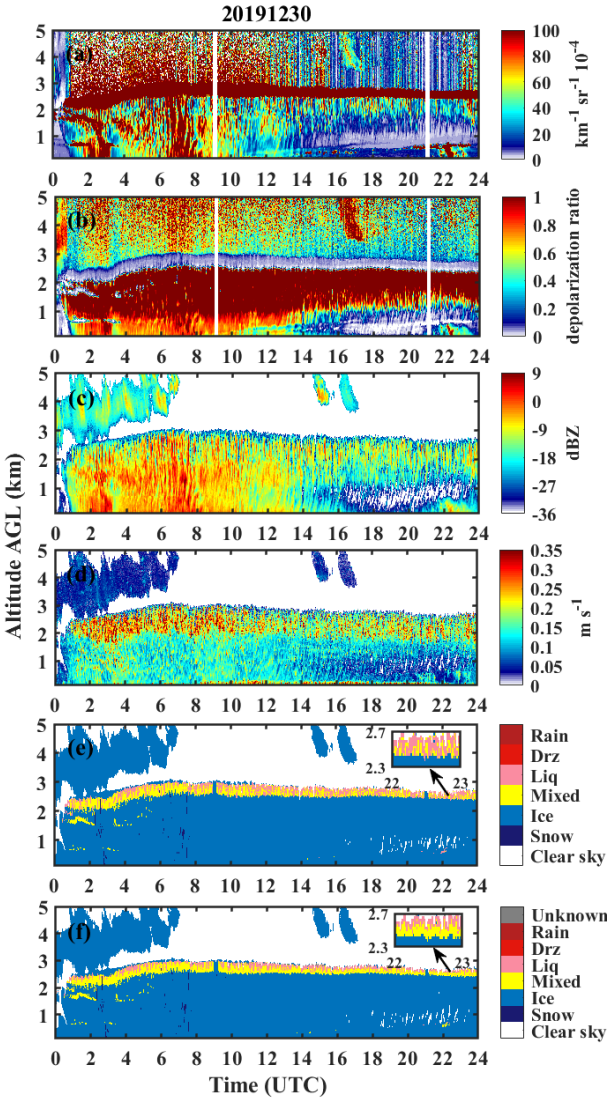


Fig. 9. Observation results of HSRL and KAZR during MOSAiC on December 30, 2019. (a) and (b) Distribution of backscattering coefficient and depolarization ratio. (c) and (d) Distribution of radar reflectivity and SW. (e) and (f) Cloud phase classification results of Shupe's method and the proposed method.

cloud, with an average N_{liq} of 35.59 cm^{-3}, whereas the average N_{liq} for liquid cloud is 75.63 cm^{-3}. This may be due to the presence of small ice crystals at lower concentrations in MPC. These small ice crystals do not significantly impact reflectivity, resulting in the reflectivity still below −17 dBZ. Hence, Shupe's method classifies Case 1 as liquid cloud. In Case 2, it shows low concentration droplets, yet this portion of cloud pixels corresponds to larger mean Doppler velocities. Thus, the proposed method identifies it as liquid cloud, while Shupe's method identifies it as MPC. Compared with Shupe's method, the algorithm proposed in this article can more accurately identify the phase of Arctic clouds and its classification results are less affected by environmental factors.

C. Case Study

To further compare the results of two cloud phase classification algorithms, we provide an example of December 30, 2019 in Fig. 9 for illustration. It can be seen that there

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are stratiform clouds within a range of 2–3 km, accompanied by a large backscattering coefficient and a small depolarization ratio. However, there is no significant increase in radar reflectivity. At the same time, an enhanced SW appears in the stratiform cloud region. This is a typical feature of polar stratiform MPCs [61], and we can also observe the MPCs are persistent.

The classification results corresponding to the proposed method and Shupe's method are generally consistent. However, compared with Shupe's method, the proposed method identifies the liquid cloud thinner and typically locates it at the cloud top, while identifying the MPC thicker below the liquid cloud. Fig. 9(e) and (f) further presents the locally magnified results of cloud phase distribution between 2200 and 2300 Coordinated Universal Time (UTC). Based on the inversion results of cloud microphysical parameters in Section IV-B, r_{eff} at the bottom of the MPC is significantly larger, with an average r_{eff} of 17.43 μm. Nevertheless, the average r_{eff} at the cloud top is 11.21 μm. Meanwhile, the proposed method obtains a more continuous cloud phase. In contrast, Shupe's method intermittently identifies the entire stratiform cloud as a liquid cloud.

V. Conclusion

This article proposes a novel Arctic cloud phase classification algorithm based on KAZR, HSRL, MWR, and temperature soundings. By combining complementary information between instruments, observed cloud pixels are classified into clear sky, aerosol, ice, mixed phase, liquid, drizzle, rain, and snow. Compared with Shupe's method, both the methods have a good consistency rate in distinguishing liquid and MPC, about 85.7%. The results indicate that misclassifying MPC as liquid cloud can significantly impact the droplet microphysical retrievals, thus it is necessary to separate the contribution of ice crystals to reflectivity when retrieving MPC microphysical parameters. Furthermore, the proposed algorithm can more sensitively identify MPCs with low concentrations of ice crystals and is less influenced by environmental factors. The resulting cloud phase classification mask can be used for ground-based instruments to correctly apply MPC microphysical retrievals. The classifier is determined based on the distribution range of droplet microphysical parameters observed by *Polar 5*. Similar methods can also be applied to classify cloud phase at other sites, but the microphysical parameters distribution observed by aircraft is required. Meanwhile, the proposed algorithm has difficulty accurately retrieving the phase of deep cloud layers and is more inclined toward identifying stratiform clouds with smaller optical depth rather than convective clouds. When the *Polarstern* experiences large roll and pitch angles, the radar beam deviates from a completely vertical orientation. This deviation, coupled with substantial horizontal wind shear, can lead to variations in SW, thereby impacting the accuracy of cloud phase classification. Consequently, our future plan is to implement the proposed algorithm at stable platform sites in the Arctic (such as the NSA site) to further validate its accuracy. In the future, we will also take into account the potential impact of molten particles, such as melting layers, which may be present at temperatures above 0 °C on the algorithm.

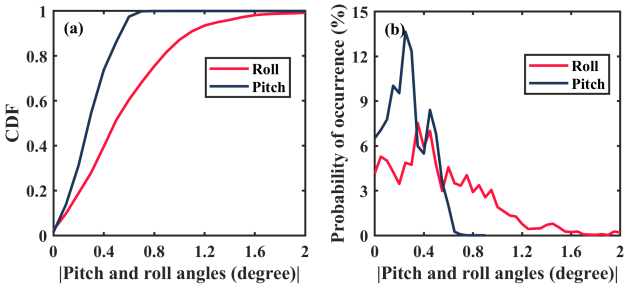


Fig. 10. (a) CDF and (b) probability of occurrence of *Polarstern*'s absolute roll and pitch angles when HSRL signal is extinguished before reaching the top of cloud.

Appendix

High concentration droplets in clouds can cause significant attenuation of HSRL signals. Statistical analysis indicates that during the MOSAiC expedition, approximately 53% of HSRL signals are able to fully penetrate the liquid-containing cloud. Meanwhile, approximately 47% of HSRL signals are extinguished before reaching the top of cloud. The proposed method uses reflectivity of KAZR and extinction coefficients of HSRL to differentiate between liquid clouds and MPCs when HSRL signal is not entirely attenuated [Fig. 2(b)]. In this instance, the Doppler information does not contribute to the cloud phase classification algorithm. Therefore, ship shaking has almost no impact on cloud classification results. For the case that 47% of HSRL signals are extinguished before reaching the top of cloud, we perform a statistical analysis of the ship's rocking angles. The (a) cumulative distribution function (cdf) and (b) probability of occurrence of *Polarstern*'s absolute roll and pitch angles are presented in Fig. 10. Overall, pitch angles tend to be relatively small, with nearly all the measurements falling below 0.7°. Meanwhile, over 60% of roll angle measurements correspond to absolute values less than 0.5°, while only about 10% exceed 1° and 2°. According to statistics, the average absolute values for the pitch angle and roll angle are approximately 0.3° and 0.5°, respectively.

In Fig. 11(a), we compare the SW associated with the cloud phase (liquid or mixed phase) derived from radar at large (roll or pitch angles greater than 0.5°) and small (roll and pitch angles less than 0.5°) rocking angles when the HSRL signal is extinguished. The results indicated that SWs are comparable under large and small rocking angles, with the average SWs of 0.3165 and 0.3175 m/s, respectively. In Fig. 11(b), we further compare the horizontal wind change rate (defined by Equation A1) associated with liquid or MPC derived from radar at large and all rocking angles when the HSRL signal is extinguished.

$$S = \frac{\sqrt{(u_i - u_{i-1})^2 + (v_i - v_{i-1})^2}}{\Delta t}. \tag{A1}$$

In the equation, u_i and v_i denote the eastward and northward components of the horizontal wind at the corresponding time and height, respectively. u_{i-1} and v_{i-1} represent the eastward and northward components of the horizontal wind at the

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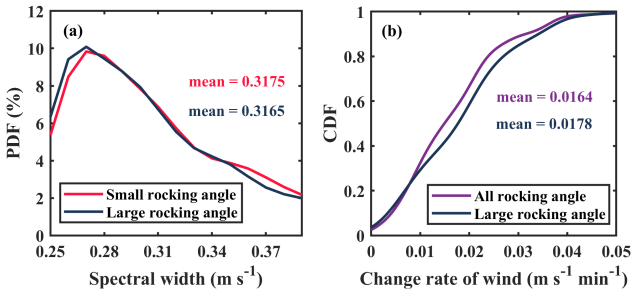


Fig. 11. (a) Probability density function (pdf) of the SW at large and small rocking angles, and (b) cdf of the horizontal wind change rate at large and all the rocking angles when the HSRL signal is extinguished.

preceding moment at the corresponding height. S signifies the change rate of horizontal wind, while Δt indicates the time interval. Due to the minimal horizontal wind change rate observed in liquid or MPCs when the HSRL signal is extinguished during large rocking angles, which is predominantly less than 0.05 m/s/min, the average wind speed change rate is calculated to be 0.0178 m/s/min. Therefore, the associated horizontal wind field is relatively uniform and exerts minimal influence on the radar SW. This leads to comparable SWs for both small and large rocking angles in Fig. 11(a). During the MOSAiC period, the ship's rocking angles remain relatively small. Minor variations in SW induced by *Polarstern*'s rocking motion do not significantly affect the overall results of cloud classification. However, it is important to note that when radar beam is not vertical pointing, substantial horizontal wind shear can lead to obvious deviations in SW, thereby impacting the outcomes of cloud phase classification.

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